Mohammad Nasr

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Robotics PhD candidate building the trust layer for physical AI: agentic architectures in which independent watchdog, judge and orchestrator agents check what a robot is doing and step in when its behavior drifts from the mission. Before that I worked on nonlinear dynamics and vibration, developing a system for autonomous testing of vibrating systems that makes characterizing nonlinear, real world dynamical behavior far more tractable. In acoustics, I study binaural hearing in animals to design bioinspired robotic listening. For me, working across disciplines isn't a detour, it's how I think.

EDUCATION

Ph.D., Robotics & Autonomous Systems
Arizona State University · Expected 2028
Fulton Fellow

M.Sc., Mechanical Engineering
Univ. of Nebraska–Lincoln · 2024
Advisor: Keegan J. Moore, Ph.D. (Dynamics & Vibration)

B.Sc., Mechanical Engineering
University of Tabriz · 2021
Advisor: Mehran Mahboubkhah, Ph.D.

TECHNICAL TOOLKIT

AGENTIC AI
Multi Agent Architectures, LLM Agents, Runtime Verification & Monitoring, Model Context Protocol (MCP)

ROBOTICS
ROS / ROS 2, MoveIt, SLAM, Motion Planning, Manipulation, Sensor Fusion

ML / AI
PyTorch, Transformers, Vision Language Models, LLMs, Reinforcement Learning, Evaluation Pipelines

SIMULATION
Gazebo, MuJoCo, Isaac Sim, MATLAB/Simulink, Digital Twins, Benchmarking

CONTROL / MATH
Optimization, System Identification, State Estimation, Numerical Methods, Linear Algebra, Statistics

PROGRAMMING
Python, C++, MATLAB, Linux, Git, Data Analysis

RESEARCH OUTPUT

2 journal publications · 2 conference presentations · 1 manuscript in preparation (see p.2)

"Smart Automatic Modal Hammer," J. Vib. Eng. Technol. (2024)

RESEARCH EXPERIENCE

Agentic Verification for Robotic Systems — Current · PhD research
Arizona State University

- Designing a hierarchical multi agent architecture (watchdog, judge, orchestrator) that independently verifies robot behavior and intervenes when actions drift from mission intent, a trust layer for physical AI.
- Separated fast watchdog monitoring from slower foundation model reasoning, so verification never blocks real time control; each agent boundary is justified by actor/checker independence, timescale separation or a certifiable authority limit.
- Built a working prototype in which agents act through tools exposed over the Model Context Protocol (MCP); paper in preparation for IEEE Robotics and Automation Letters (RA-L).

ACADEMIC PROJECTS

Active Perception for Object Localization — Jan–May 2026
Arizona State University

- Built a TurtleBot4 active perception system pairing next best view selection with a confidence scoring algorithm to improve pose estimation reliability.
- Integrated Nav2 global/local planning with an ORB SLAM3 + EKF + custom A* fallback for mapless navigation.

Acoustic Based, Bioinspired Robotic Platforms Design — Aug 2024–Dec 2025
Arizona State University

- Analyzed hearing systems of domestic cats, arctic fox, and aye aye to design bioinspired robotic hearing methods.
- Coauthored "Guided by Hearing," accepted in the Journal of Material Evaluations.

Assistive Follower Robot (Load Bearing) — Jan–May 2025
Arizona State University

- Built a TurtleBot 4 follower robot with depth camera, LiDAR, and ROS 2 for hands free tracking and obstacle avoidance.
- Trained a lightweight ResNet18 model for real time shoe detection on Raspberry Pi.

Solving a 4×4 Maze with MyCobot Pro 600 — Aug–Dec 2024
Arizona State University

- Implemented the A* algorithm for maze solving and robot motion planning, verified via digital twin simulation.

JOURNAL PUBLICATIONS

Nasr, M., Masurkar, N., Meshkizadeh, P., & Dehghan-Niri, E. (2025). Guided by Hearing: Robotic Motion Planning for Precision in Gas Leakage Localization. *Journal of Material Evaluations.* Accepted for publication.

Nasr, M., Singh, A., & Moore, K.J. Smart Automatic Modal Hammer: Predictor–Corrector Approach for Accurate Excitation of Dynamical Systems. *J. Vib. Eng. Technol.* 12 (Suppl 1), 1045–1063 (2024). doi.org/10.1007/s42417-024-01461-6

In preparation: Nasr, M. Watchdog: Physically Grounded Conformance Monitoring for Robotic Agentic AI. In preparation for submission to *IEEE Robotics and Automation Letters (RA-L).*

MORE PROJECTS

Autonomous Testing of Vibrating Systems
Univ. of Nebraska–Lincoln · Aug 2022–May 2024

RL for Robot Navigation (Q Learning)
Univ. of Nebraska–Lincoln · Jan–May 2024

Autonomous Navigation, F1Tenth Platform
Univ. of Nebraska–Lincoln · Jan–May 2024

Robotic Arm Soccer Player
Univ. of Nebraska–Lincoln · Aug–Dec 2023

Planar Circular Restricted Three Body Problem
Univ. of Nebraska–Lincoln · Jan–May 2023

AVIATION TRAINING

B1, Airframe Mechanics & Aircraft Maintenance Technology
Civil Aviation Technology College (CATC), Tehran · Sep 2020–Jul 2021

Passed civil aircraft maintenance coursework with EASA modules covering aerodynamics, structures, gas turbine engines and propeller systems.

LEADERSHIP

Local Committee Vice President
AIESEC, Tabriz · May–Sep 2017

Built volunteering projects that brought international students to Iran; negotiated financial and accommodation support with local partners.

CONFERENCE PRESENTATIONS

Nasr, M., Masurkar, N., Meshkizadeh, P., & Dehghan-Niri, E. "Gas Leakage Localization Through Bioinspired Motion Strategy on Robotic Platforms," ASNT Research Symposium 2025, Indiana, June 23–27.

Nasr, M., Singh, A., & Moore, K.J. "Smart Automatic Modal Hammer for Studying Nonlinear Dynamical Systems," International Modal Analysis Conference XLII, Orlando, FL, Jan 29–Feb 1, 2024.

TEACHING EXPERIENCE

Teaching Assistant, Robotic Systems 1 Jan 2026–Present
Dept. of Manufacturing & Systems, Arizona State University

- Conducted lab sessions using Dobot robotic arms; designed lab exercises and final project assignments.
- Graded student homework assignments and lab reports.

Teaching Assistant, Engineering Dynamics Aug–May 2024
Mechanical & Materials Engineering, Univ. of Nebraska–Lincoln

- Conducted recitation sessions; coached students on course content and problem solving procedures.
- Corrected and graded student homework problems and exams.

TA, Kinematics & Dynamics of Machinery; Elements of Machine Design Aug 2022–May 2023
Mechanical & Materials Engineering, Univ. of Nebraska–Lincoln

- Addressed student concerns on course materials and solutions; corrected and graded homework and exams.

PROFESSIONAL EXPERIENCE

Cofounder & Chief Product Designer May–Sep 2021
VISION Company, Tabriz, Iran

- Designed automatic hand and foot disinfectant devices using SolidWorks, adopted in schools, restaurants, and stores.
- Programmed Arduino Uno as the main control unit for the automation process.
- Generated nearly **30%** revenue through product sales.
- Designed an automatic face mask delivery machine for malls, bus stops, and train stations.

Aerial Robotics Team Leader; Aerodynamics & Structural Design Lead Dec 2017–Jun 2019
Robotic Lab, University of Tabriz, Iran

- Led the aerial robotics team for the Iran Open Robotics competition: designed the quadrotor frame, flew as pilot, and secured financial and lab support from research center supervisors.
- Designed fuselage, wings and tail in SolidWorks, analyzed aerodynamics in ANSYS, and built the first balsa wood prototype for the national model airplane competition.
- Optimized an airfoil into a tapered wing, later built by 3D printing and balsa wood for the wind tunnel teaching lab.